Optimization

General Information

Welcome to COMP9602. This core research course will provide you the theory, algorithms, and applications of convex optimization. The course consists primarily of lectures. The assessment consists of 3 assignments, 1 project, and 1 final examination.

Main topics to be discussed include:

On the theory of convex optimization: convex set and functions, linear programming, quadratic programming, semidefinite programming, geometric programming, integer programming, vector optimization, duality theory (dual, Lagrange multiplier, KKT conditions), etc.

On algorithms to solve convex optimization problems: gradient descent algorithm, Newton's method, interior point method, ellipsoid method, subgradient algorithm, decomposition methods, etc.

We will touch upon non-convex optimization briefly as well.

Learning Outcomes

On successful completion of the course, students should:

CLO1 [**Basic Concepts**]**:** be able to master the key concepts on convex sets, convex functions and convex programs.

CLO2 [**Key Theory**]**:** be able to master the key theory on duality.

CLO3 [**Representative Algorithms**]**:** be able to master the representative algorithms to solve convex programs.

CLO4 [**Problem Solving**]**:** be able to model practical problems which they identify into convex programs, design/apply algorithms to solve them and present them in the format of academic papers.

Teaching Staff

Teacher: Prof. Chuan Wu
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Course Contact

Course available on Moodle: http://moodle.hku.hk

Text and Reference

Main text:
Stephen Boyd and Lieven Vandenberghe, **Convex Optimization**, Cambridge University Press, 2004. (pdf available online https://web.stanford.edu/~boyd/cvxbook/)

Reference:
(1) Christos H. Papadimitriou and Kenneth Steiglitz, **Combinatorial Optimization: Algorithms and Complexity**, Dover Publications, 1998.
(2) Dimitri P. Bertsekas, **Convex Optimization Theory**, Athena Scientific, 2009 (http://web.mit.edu/dimitrib/www/Convex_Theory_Entire_Book.pdf)
(3) Dimitri P. Bertsekas, **Convex Optimization Algorithms**, Athena Scientific, 2015 (http://web.mit.edu/dimitrib/www/Contents_Preface_NEW_ALG.pdf)
(4) Dimitri P. Bertsekas, **Nonlinear Programming (3nd edition)**, Athena Scientific, 2016

Assessment

The assessment of the course includes 3 assignments, 1 project, and the final examination. The composition of the final grade is as follows:

Item	Weight
Assignments	36%
Project	24%
Final Exam	40%

Time and Venue

Time	Venue
Tuesdays 10:00 11:20am	CYCP1 (LG1 Chong Yuet Ming Chemistry Building)
Thursdays 10:00 11:20am	MWT5 (1/F Meng Wah Complex)

Lecture Schedule

Tentative schedule: (readings refer to chapters in the main text unless stated otherwise)

Week	Date	Lecture Topic	Reading	Notes
1	Sep. 1	Introduction to convex optimization	Chapter 1	
	Sep. 3	Convex set	Chapter 2	
2	Sep. 8	Convex function	Chapter 3	
	Sep. 10	Convex function	Chapter 3	Release assignment 1
3	Sep. 15	Convex function; standard forms of optimization problems, convex program	Chapters 3, 4	
	Sep. 17	Equivalent programs	Chapter 4	
4	Sep. 22	Quasi-convex program	Chapter 4	
	Sep. 24	Linear program, integer linear program	Chapter 4	
5	Sep. 29	Quadratic program, second-order cone programming	Chapter 4	
	Oct. 1	Public holiday		
6	Oct. 6	Geometric program, semidefinite program, vector optimization	Chapter 4	
	Oct. 8	Duality	Chapter 5	
7 (Reading Week)	Oct. 13	Duality	Chapter 5	
	Oct. 15	Duality	Chapter 5	Release assignment 2
8	Oct. 20	Duality	Chapter 5	
	Oct. 22	Algorithms for unconstrained optimization	Chapter 9	
9	Oct. 27	Algorithms for equality constrained optimization	Chapter 10	
	Oct. 29	Interior-point methods	Chapter 11	
10	Nov. 3	Interior-point methods	Chapter 11	

Week	Date	Lecture Topic	Reading	Notes
	Nov. 5	Subgradient methods	Reference 5-6	
11	Nov. 10	Localization methods	Reference 7	Release assignment 3
	Nov. 12	Decomposition methods	Reference 8	
12	Nov. 17	Decomposition methods; branch and bound methods	References 8, 9	
	Nov. 19	Other methods for nonconvex optimization		